

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250271082

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Reyes Garcia; Jesus et al.

ASSEMBLY FOR ENSURING ATTACHMENT

Abstract

Attachment assemblies, vehicles with attachments assemblies, and methods for ensuring attachment are disclosed. A method includes providing an attachment assembly in an initial configuration. The attachment assembly includes a first component having a scannable feature and a second component having a shield portion and a window. In the initial configuration, the scannable feature is received within the second component and the shield portion covers the scannable feature. The method further includes advancing the attachment assembly in a first direction into contact with an attachment feature such that the first component abuts the attachment feature while the second component continues moving in the first direction relative to the first component until the attachment assembly is in an attached configuration in which the scannable feature is revealed by the window; and scanning the scannable feature through the window with a scanner to confirm that the attachment assembly has been attached.

Inventors: Reyes Garcia; Jesus (Zinacantepec, MX), Velez; Julio Alberto (Toluca de Lerdo, MX), Fernandez Davila; Gabriel (Lerma, MX), Cortes Fabela; Sergio (Temixco, MX)

Applicant: GM GLOBAL TECHNOLOGY OPERATIONS LLC (Detroit, MI)

Family ID: 1000007722813

Assignee: GM GLOBAL TECHNOLOGY OPERATIONS LLC (Detroit, MI)

Appl. No.: 18/587312

Filed: February 26, 2024

Publication Classification

Int. Cl.: F16L3/12 (20060101); B60K11/02 (20060101)

U.S. Cl.:

Background/Summary

INTRODUCTION

[0001] The disclosure relates to attachment features for vehicles, and may in particular relate to assemblies for ensuring proper attachment between two components.

[0002] The battery is the main energy source of an electric vehicle (EV) that supplies the power to propel the vehicle and to power other auxiliary systems. The performance of the battery is greatly influenced by its operating temperature. For example, certain battery types are required to operate at a temperature of from 20 to 40° C. To cool the battery for proper operation, a cooling system must maintain a proper flow rate and pressure of coolant at the inlet and outlet of the battery.

[0003] A coolant hose is used to deliver and remove coolant from the battery. The coolant hose is secured to the vehicle to prevent improper positioning, kinks, or damage to the hose. Therefore, attachment of the hose within the vehicle is important to maintain effective cooling of the battery and proper operation of the vehicle.

[0004] Accordingly, it is desirable to provide new and improved methods and assemblies for ensuring attachment of vehicle components, such as coolant hoses. Furthermore, other desirable features and characteristics of the present disclosure will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and the foregoing technical field and background.

SUMMARY

[0005] In an embodiment, a method for ensuring attachment is provided. The method includes providing an attachment assembly in an initial configuration. The attachment assembly includes a first component having a scannable feature; and a second component having a shield portion and a window. In the initial configuration, the scannable feature is received within the second component and the shield portion covers the scannable feature. The method further includes advancing the attachment assembly in a first direction into contact with an attachment feature such that the first component abuts the attachment feature while the second component continues moving in the first direction relative to the first component until the attachment assembly is in an attached configuration in which the scannable feature is revealed by the window; and scanning the scannable feature through the window with a scanner to confirm that the attachment assembly has been attached to the attachment feature.

[0006] In certain embodiments, the method further includes communicating a signal from the scanner indicating that the attachment assembly has been attached to the attachment feature.

[0007] In certain embodiments of the method, the second component includes a body portion connected to a ring portion, and the method further includes surrounding a hose with the ring portion.

[0008] In certain embodiments of the method, the second component includes a body portion connected to a ring portion, and the method further includes surrounding a hose with the ring portion before advancing the attachment assembly in the first direction into contact with the attachment feature and before scanning the scannable feature through the window with the scanner to confirm that the attachment assembly has been attached to the attachment feature.

[0009] In certain embodiments of the method, the first component includes a first locking feature; the second component includes a second locking feature; and the method includes locking the attachment assembly in the attached configuration by engaging the first locking feature and the second locking feature.

[0010] In certain embodiments of the method, the first component includes a first retaining feature;

the second component includes a second retaining feature; and the method includes retaining the attachment assembly in the initial configuration by engaging the first retaining feature and the second retaining feature until a force from advancing the attachment assembly in the first direction while the first component abuts the attachment feature causes the first retaining feature and the second retaining feature to disengage.

[0011] In certain embodiments of the method, the attachment feature includes a surface formed with a hole; the second component includes a distal projection; advancing the attachment assembly in the first direction includes inserting the distal projection into the hole; advancing the attachment assembly in the first direction is continued until the shield portion abuts the surface; the attachment assembly is in the attached configuration when the shield portion abuts the surface; and while advancing the attachment assembly in the first direction the first component abuts the surface.

[0012] In certain embodiments of the method, the surface is a first surface of a wall; the wall has an opposite second surface; the hole extends through the wall; the distal projection includes a deformable radially-outward extension; advancing the attachment assembly in the first direction including deforming the deformable radially-outward extension to pass through the hole; and the deformable radially-outward extension extends outward after passing through the hole and prevents removal of the distal projection from the hole.

[0013] In certain embodiments of the method, the attachment feature includes a stud extending from a surface; the second component includes an inward-extending arm; advancing the attachment assembly in the first direction includes contacting the inward-extending arm with the stud; advancing the attachment assembly in the first direction is continued until the shield portion abuts the surface; and the attachment assembly is in the attached configuration when the shield portion abuts the surface.

[0014] In certain embodiments of the method, the inward-extending arm includes a plurality of inward-extending arms that define a central passageway; and while advancing the attachment assembly in the first direction a distal portion of the stud extends through the central passageway.

[0015] In another embodiment, an attachment assembly is provided and includes a first component having a scannable feature; and a second component having a shield portion and a window. The first component is movable relative to the second component in a first direction from an initial configuration to an attached configuration. In the initial configuration the scannable feature is received within the second component and the shield portion covers the scannable feature. In the attached configuration, the scannable feature is revealed by the window.

[0016] In certain embodiments of the attachment assembly, the first component includes a first locking feature; the second component includes a second locking feature; and the first locking feature is configured to engage the second locking feature to lock the attachment assembly in the attached configuration.

[0017] In certain embodiments of the attachment assembly, the first component includes a first retaining feature; the second component includes a second retaining feature; and the first retaining feature is configured to hold the second retaining feature in the initial configuration and to release the second retaining feature upon application of a force in the first direction.

[0018] In certain embodiments of the attachment assembly, the second component includes a distal projection extending in a second direction opposite the first direction and configured for insertion in a hole in a wall; and the distal projection includes a deformable radially-outward extension configured to engage a backside of the wall.

[0019] In certain embodiments of the attachment assembly, the second component includes inward-extending arms that terminate at a central passageway.

[0020] In another embodiment, a vehicle is provided and includes a coolant system including a hose; and an attachment assembly including a first component having a scannable feature and a second component including a ring portion engaging the hose and having a shield portion and a window. The first component is movable relative to the second component in a first direction from

an initial configuration to an attached configuration. In the initial configuration the scannable feature is received within the second component and the shield portion covers the scannable feature. In the attached configuration, the scannable feature is revealed by the window.

[0021] In certain embodiments, the vehicle further includes a wall formed with a hole extending from a first surface of the wall to an opposite second surface of the wall, the second component includes a distal projection extending through the hole, the distal projection includes a radially-outward extension having a diameter greater than a diameter of the hole, the shield portion abuts the first surface of the wall, and the first component abuts the first surface of the wall.

[0022] In certain embodiments, the vehicle further includes a stud extending from a surface, the second component includes inward-extending arms that terminate at a central passageway, the stud passes through the central passageway, and the first component abuts the surface.

[0023] In certain embodiments of the vehicle, the first component includes a first locking feature, the second component includes a second locking feature, and the first locking feature is configured to engage the second locking feature to lock the attachment assembly in the attached configuration.

[0024] In certain embodiments of the vehicle, the first component includes a first retaining feature, the second component includes a second retaining feature, and the first retaining feature is configured to hold the second retaining feature in the initial configuration and to release the second retaining feature upon application of a force in the first direction.

Description

DESCRIPTION OF THE DRAWINGS

[0025] The present disclosure will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:

[0026] FIG. 1 is a schematic layout of a vehicle having a cooling system in accordance with exemplary embodiments;

[0027] FIG. 2 is a perspective view schematic of the attachment of a coolant hose of FIG. 1 with an attachment assembly in accordance with exemplary embodiments;

[0028] FIG. 3 is a perspective view schematic of a portion of a ring portion of the attachment assembly of FIG. 2 in accordance with certain embodiments;

[0029] FIG. 4 is a perspective view schematic of an outer member of the attachment assembly of FIG. 2 in accordance with certain embodiments;

[0030] FIG. 5 is a perspective view schematic of an inner member of the attachment assembly of FIG. 2 in accordance with certain embodiments;

[0031] FIG. 6 is a side view schematic of the attachment assembly of FIG. 2 in an initial configuration before contact with an attachment feature in accordance with certain embodiments;

[0032] FIG. 7 is a side view schematic of the attachment assembly of FIG. 2 in an attached configuration and in engagement with an attachment feature in accordance with certain embodiments;

[0033] FIG. 8 is a perspective view schematic of the attachment of a coolant hose of FIG. 1 with an attachment assembly in accordance with exemplary embodiments;

[0034] FIG. 9 is a perspective view schematic of a portion of a ring portion and fastener member of the attachment assembly of FIG. 8 in accordance with certain embodiments;

[0035] FIG. 10 is a perspective view schematic of an outer member of the attachment assembly of FIG. 8 in accordance with certain embodiments;

[0036] FIG. 11 is a perspective view schematic of an inner member of the attachment assembly of FIG. 8 in accordance with certain embodiments;

[0037] FIG. 12 is a side view schematic of the attachment assembly of FIG. 8 in an initial configuration before contact with an attachment feature in accordance with certain embodiments;

[0038] FIG. 13 is a side view schematic of the attachment assembly of FIG. 8 in an initial configuration upon contact with an attachment feature in accordance with certain embodiments; [0039] FIG. 14 is a side view schematic of the attachment assembly of FIG. 8 in an attached configuration and in engagement with an attachment feature in accordance with certain embodiments, and [0040] FIG. 15 is a flow chart illustrating a method for ensuring attachment in accordance with certain embodiments.

DETAILED DESCRIPTION

[0041] The following detailed description is merely exemplary in nature and is not intended to limit the application and uses of embodiments herein. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding introduction, brief summary or the following detailed description. As used herein, the term “module” refers to any hardware, software, firmware, electronic control unit or component, processing logic, and/or processor device, individually or in any combination, including without limitation: application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality.

[0042] Embodiments of the present disclosure may be described herein in terms of functional and/or logical block components and various processing steps. It should be appreciated that such block components may be realized by any number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of the present disclosure may employ various integrated circuit components, e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like, which may conduct a variety of functions under the control of one or more microprocessors or other control devices. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with any number of automated driving systems including cruise control systems, automated driver assistance systems and autonomous driving systems, and that the vehicle system described herein is merely one example embodiment of the present disclosure.

[0043] For the sake of brevity, conventional techniques related to signal processing, data transmission, signaling, control, and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure.

[0044] Embodiments herein provide for ensuring attachment between components, such as in a vehicle. In certain embodiments, a hose is encircled or surrounded by a ring member of an attachment assembly. The attachment assembly further includes components configured to be mounted to an attachment feature of the vehicle. The attachment assembly includes a scannable feature that may be scanned after the attachment assembly is properly installed and connects the components. For example, the attachment assembly has an initial configuration that covers or hides the scannable feature. The attachment assembly further has an attached configuration in which the assembly uncovers or reveals the scannable feature. The attachment assembly may self-lock itself in the attached configuration. As a result, proper connection between two components may be evaluated by reviewing scanned scannable features.

[0045] With reference to FIG. 1, certain features of a vehicle 10 are illustrated in functional block diagram form. In certain examples, the vehicle 10 comprises an automobile. In various examples, the vehicle 10 may be any one of a number of different types of automobiles, such as, for example, a sedan, a wagon, a truck, or a sport utility vehicle (SUV), and may be two-wheel drive (2WD) (i.e., rear-wheel drive or front-wheel drive), four-wheel drive (4WD), or all-wheel drive (AWD).

The vehicle **10** may also incorporate any one of, or combination of, a number of different types of engines, such as, for example, a gasoline or diesel fueled combustion engine, a “flex fuel vehicle” (FFV) engine (i.e., using a mixture of gasoline and alcohol), a gaseous compound (e.g., hydrogen and/or natural gas) fueled engine, a combustion/electric motor hybrid engine (i.e., such as in a hybrid electric vehicle (HEV)), and an electric motor.

[0046] As depicted in FIG. **1**, the exemplary vehicle **10** generally includes a chassis **12**, a body **14**, four wheels **16**, and an electronic control system **18**. The body **14** is arranged on the chassis **12** and substantially encloses the other components of the vehicle **10**. The body **14** and the chassis **12** may jointly form a frame. The wheels **16** are each rotationally coupled to the chassis **12** near a respective corner of the body **14**.

[0047] In the exemplary embodiment illustrated in FIG. **1**, the vehicle **10** includes an actuator assembly **20**, a battery (or a DC power supply) **22**, and a power converter assembly (e.g., an inverter or inverter assembly) **24**. The actuator assembly **20** may include an electric motor/generator (or motor) **26** and/or a combustion engine **28**.

[0048] Still referring to FIG. **1**, the electric motor **26** and the combustion engine **28** are integrated such that one or both are mechanically coupled to at least some of the wheels **16** through one or more drive shafts **29**.

[0049] The vehicle **10** further includes a thermal management system **30** including a radiator **32** and coolant circulation hoses **34**. The radiator **32** may be connected to the frame at an outer portion thereof. The coolant circulation hoses **34** may provide multiple cooling channels that contain a cooling fluid (i.e., coolant) such as water and/or ethylene glycol (i.e., “antifreeze”). The coolant circulation hoses **34** may be coupled to the combustion engine **28** and/or the inverter **24**. In certain embodiments, these components control the temperature of the battery **22**, inverter **24**, and motor **26** to provide optimal performance.

[0050] Referring again to FIG. **1**, in the depicted embodiment, the inverter **24** receives and shares coolant with the electric motor **26**. However, other embodiments may use separate coolants for the inverter **24** and the electric motor **26**. The radiator **32** may be similarly connected to the inverter **24** and/or the electric motor **26**.

[0051] The electronic control system **18** is in operable communication with the actuator assembly **20**, the high voltage battery **22**, and the inverter **24**. Although not shown in detail, the electronic control system **18** includes various sensors and automotive control modules, or electronic control units (ECUs), such as an inverter control module, a motor controller, and a vehicle controller, and at least one processor and/or a memory.

[0052] In electric vehicles (EVs) **10**, the failure to appropriately circulate coolant may represent an overheating situation that may lead to a loss of propulsion during vehicle operation. Therefore, a high confidence attachment may be used to prevent the coolant circulation hose **34** from becoming loose and being punctured or otherwise forming a leak.

[0053] Referring to FIG. **2**, an exemplary coolant circulation hose **34** of the thermal management system **30** is shown mounted to a portion of the vehicle **10**. Specifically, in the illustrated embodiment, the hose **34** is mounted to an attachment feature **42** on a component **44** of the vehicle **10**.

[0054] As shown in FIG. **2**, an attachment assembly **100** is used to mount the coolant circulation hose **34** to the attachment feature **42**. The attachment assembly **100** may include multiple portions or components, such as a ring member **200**, a base or outer member **300**, and an inner member **400**, that are connected together to provide a method for ensuring that the coolant circulation hose **34** is properly mounted to the component **44** of the vehicle **10**. In FIG. **2**, the attachment assembly **100** is shown in an attached configuration **100''**.

[0055] FIGS. **3-5** separately illustrate each member **200**, **300**, and **400** of the attachment assembly **100** of FIG. **2**, respectively.

[0056] In FIG. **3**, a proximal portion of the ring member **200** is shown. As shown, an opening **202**

is formed at a proximal end **204** of the ring member **200**. The ring member **200** further includes a flexible body **206** that extends in a distal direction away from the proximal end **204** to a distal end (not shown). In FIG. 3, the ring member **200** is shown in an un-coupled configuration in which the flexible body **206** is not received within the opening **202**. To engage a coolant circulation hose **34**, the flexible body **206** may be looped around the hose **34** and the distal end thereof may be passed through the opening **202**. In exemplary embodiments, the opening **202** and flexible body **206** are formed with cooperating structures to allow the flexible body **206** to be inserted into the opening **202** while preventing the flexible body **206** from being withdrawn from the opening **202**.

[0057] In FIG. 4, an exemplary embodiment of the outer member **300** is illustrated. As shown, the outer member **300** includes a main portion **310** that extends from a bottom end **302** to a top end **304**. The main portion **310** is annular and defines a vertically extending inner channel **320** in communication with a top opening **322** at the top end **304**.

[0058] Adjacent to the top end **304**, a pair of aligned slots **324** are provided to receive the flexible body **206** of the ring member **200** to allow attachment of the ring member **200** to the outer member **300**. As shown, the outer member **300** may include an alternate pair of aligned slots offset by ninety degrees to allow the ring member **200** to be attached to the outer member **300** at a desired orientation.

[0059] The outer member **300** may be provided with a pair of opposite retaining features **332**, such as openings. Further, the outer member **300** may be provided with a pair of opposite locking features **334**, such as openings. In the illustrated embodiment, the retaining features **332** and locking features **334** are vertically aligned, with the retaining features **332** being located between the locking features **334** and the bottom end **302**.

[0060] As shown, the outer member **300** further includes a window compartment **340** joined to the main portion **310**. The window compartment **340** has an outer wall **342** that defines a vertically-extending inner channel **344**. The inner channel **344** is covered by a shield portion **346** of the outer wall **342**. Further, a window **348** is formed through the outer wall **342** and is open to the inner channel **344**.

[0061] FIG. 5 illustrates an exemplary inner member **400**. As shown, the inner member **400** may include vertically extending posts **410** that are connected to an annular body portion **420**. Exemplary vertically extending posts **410** have some flexibility to allow deflection as described below.

[0062] Further, the inner member **400** may include a face portion **430** that is connected to the annular body portion **420** by a radially extending member (hidden in FIG. 5). The face portion **430** includes an outer surface **432** on which a scannable feature **434**, such as indicia or a QR code, may be located.

[0063] As shown, the inner member **400** may include radially inward extending arms **422** connected to the annular body portion **420**. The arms **422** may be circumferentially spaced on the annular body portion **420**. Further, the arms **422** may terminate at distal ends **424** that define a central passageway **426** extending vertically through the inner member **400**.

[0064] As shown in FIG. 5, the vertically extending posts **410** have a bottom end **412** that is connected to the annular body portion **420**. The vertically extending posts **410** extend upward from the bottom end **412** to a top end **414**. At the top end **414**, the posts **410** are formed with radially outward extending projections **416**. Each radially outward extending projection **416** has a sloping surface **418**.

[0065] Cross-referencing FIGS. 3-5, it may be seen that the attachment assembly **100** of FIG. 2 may be assembled by receiving the inner member **400** within the outer member **300** and connecting the ring member **200** to the outer member **300**.

[0066] For example, the posts **410** and annular body portion **420** are received within the vertically extending inner channel **320** in the main portion **310** of the outer member **300**. Further, the face portion **430** of the inner member **400** is received within the vertically-extending inner channel **344**

of the window compartment **340**.

[0067] In an initial configuration, the radially outward extending projections **416** of the posts **410** are received in the retaining feature openings **332**. In the initial configuration, the radially outward extending projections **416** of the posts **410** engage with the retaining feature openings **332** to prevent downward movement of the inner member **400** relative to the outer member **300**, i.e., removal of the inner member **400** from the outer member **300**.

[0068] Further, in the initial configuration, the scannable feature **434** on the face portion **430** is covered by the shield portion **346** of the outer wall **342** of the window compartment **340**.

[0069] In FIG. **6**, the attachment assembly **100** is shown in the initial configuration **100'** positioned over an attachment feature **42** of the vehicle **10**. In the illustrated embodiment, the attachment feature **42** includes a stud **50** extending from a surface **46** of a component **44** of the vehicle **10**. As shown, the component **44** may be a wall and have an opposite surface **48**.

[0070] In FIG. **6**, the radially outward extending projections **416** of the inner member **400** extend through the outer member **300** and are visible. As shown, the ring member **200** encircles the coolant circulation hose **34**. As further shown, the shield portion **346** of the outer member **300** covers the scannable feature of the inner member **400** such that the scannable feature is not visible.

[0071] Further, the flexible body **206** is inserted through a pair of aligned slots **324**. The flexible body **206** may be further inserted through the opening **202** at the proximal end **204** of the ring member **200** and tightened around the coolant circulation hose **34**.

[0072] In FIG. **7**, the attachment assembly **100** is shown after a downward force has pushed the attachment assembly **100** toward the vehicle component **44**. The attachment assembly **100** is in the attached configuration **100''**. The ring member **200** is not shown for purposes of clarity.

[0073] As the attachment assembly **100** is pressed downward, the stud **50** moves upward relative to the attachment assembly **100**, and the stud **50** contacts the radially inward extending arms **422**. Upon contact, the arms **422** may deflect upward. As the outer member **300** continues movement downward, friction between the stud **50** and the arms **422** causes the inner member **400** to remain stationary or slow. As a result, the sloping surfaces **418** of the posts **410** cause the radially outward projections **416** to retract within the outer member **300** and out of engagement with the retaining feature openings **332**. The inner surface of the outer member **300** may be formed with channels that limit the radially outward projections **416** to vertical movement.

[0074] A downward force may cause the outer member **300** to continue moving downward until the outer member **300** contacts the surface **46** of the vehicle component **44**. In other embodiments, downward movement of the outer member **300** may stop before contacting the surface **46** of the vehicle component **44**.

[0075] As shown in FIG. **7**, when relative movement between the outer member **300** and inner member **400** is sufficient, the radially outward projections **416** reach the locking feature openings **334** and extend outward, into the openings **334** and through the outer member **300**. Engagement between the radially outward projections **416** and the locking feature openings **334** prevent downward movement of the inner member **400** relative to the outer member **300**. Thus, the attachment assembly **100** is locked in the attached configuration **100''**. Internal surfaces of the outer member **300** may abut the exterior surfaces of the inner member **400** in the attached configuration **100''** to prevent further upward movement of the inner member **400** relative to the outer member **300**.

[0076] In FIG. **7**, the distal end of the stud **50** extends out of the outer member **300**. In other embodiments, the distal end of the stud **50** remains within the outer member **300**.

[0077] In the attached configuration **100''**, the scannable feature **434** is aligned with, and visible through, the window **348** in the outer member **300**.

[0078] FIGS. **2-7** illustrate an embodiment of an attachment assembly **100** for use with an attachment feature **42** in the form of a projection like stud **50**. In other embodiments, the attachment assembly **100** may be used with an attachment feature **42** in the form of a hole in

surface.

[0079] For example, as shown in FIG. 8, an exemplary attachment assembly **100** is illustrated in the attached configuration **100''**. In FIG. 8, the attachment assembly **100** is attached to an attachment feature **42** in the form of a hole in a surface **46** of a vehicle component **44**. As shown, the attachment assembly **100** includes a ring member **200**, outer member **300**, and inner member **400**. The ring member **200** circumscribes a coolant circulation hose **34**.

[0080] FIGS. 9-11 separately illustrate each member **200**, **300**, and **400** of the attachment assembly **100** of FIG. 8, respectively.

[0081] In FIG. 9, the ring member **200** is shown in connection with a projection or fastener member **500**. As shown, the ring member **200** includes an opening **202** formed at a proximal end **204**. The ring member **200** further includes a flexible body **206** that extends in a distal direction away from the proximal end **204** to a distal end **208**. In FIG. 9, the ring member **200** is shown in a coupled configuration in which the distal end **208** is passed through the opening **202** and the flexible body **206** is received within the opening **202**. Typically, the ring member **200** encircles a hose **34**, which is not shown in FIG. 9. In exemplary embodiments, the opening **202** and flexible body **206** are formed with cooperating structures to allow the flexible body **206** to be inserted into the opening **202** while preventing the flexible body **206** from being withdrawn from the opening **202**.

[0082] As shown, the ring member **200** is attached to a fastener member **500**. The fastener member **500** may include a shaft portion **502** that extends distally from a head **504** to a distal end **506**. As shown, radially outward extensions **508** are formed on the shaft portion **502** near the distal end **506**. The radially outward extensions **508** may be rings around the shaft portion **502**.

[0083] The ring member **200** may be attached to the fastener member **500** by passing through an opening (not visible in FIG. 9) formed on the head **504** of the fastener member **500**. Alternatively, the proximal end **204** of the ring member **200** may be integral with, i.e., part of, the head **504** of the fastener member **500**. In other embodiments, the ring member **200** and fastener member **500** are connected in other suitable manners.

[0084] In FIG. 10, an exemplary embodiment of the outer member **300** is illustrated. As shown, the outer member **300** includes a main portion **310** that extends from a bottom end **302** to a top end **304**. The main portion **310** is annular and defines a vertically extending inner channel **320** in communication with a top opening **322** at the top end **304**.

[0085] Adjacent to the top end **304**, a pair of aligned slots **324** are provided and may receive the flexible body **206** of the ring member **200** or a portion of the head **504** of the fastener member **500** to allow attachment of the ring member **200** and fastener member **500** to the outer member **300**. As shown, the outer member **300** may include an alternate pair of aligned slots offset by ninety degrees to allow the ring member **200** and/or fastener member **500** to be attached to the outer member **300** at a desired orientation.

[0086] The outer member **300** may be provided with a pair of opposite retaining features **332**, such as openings. Further, the outer member **300** may be provided with a pair of opposite locking features **334**, such as openings. In the illustrated embodiment, the retaining features **332** and locking features **334** are vertically aligned, with the retaining features **332** being located between the locking features **334** and the bottom end **302**. In certain embodiments, a second set of locking features **334** are provided. In such an embodiment, the first set of locking features **334** may serve as retaining features in an attached configuration **100''** of the attachment assembly **100**.

[0087] As shown, the outer member **300** includes a shield portion **346** that covers the inner channel **320**. Further, a window **348** is formed in the outer member **300** and is open to the inner channel **320**.

[0088] FIG. 11 illustrates an exemplary inner member **400**. As shown, the inner member **400** may include vertically extending posts **410** that are connected to an annular body portion **420**.

[0089] Further, the inner member **400** a scannable feature **434**, such as indicia or a QR code,

located on the annular body portion **420**.

[0090] As shown in FIG. **11**, the vertically extending posts **410** extend downward from a top end **414** at the annular body portion **420** to bottom ends **412**. At the top end **414**, the posts **410** are formed with radially outward extending projections **416**. Each radially outward extending projection **416** has a sloping surface **418**. In certain embodiments, a second set of radially outward extending projections **416** are formed at the top end **444** of the inner member **400**, in vertical alignment with the lower set of radially outward extending projections **416**.

[0091] Cross-referencing FIGS. **9-11**, it may be seen that the attachment assembly **100** of FIG. **8** may be assembled by receiving the inner member **400** within the outer member **300** and connecting the ring member **200** to the outer member **300**. For example, the inner member **400** is received within the vertically extending inner channel **320** of the outer member **300**.

[0092] In an initial configuration, the radially outward extending projections **416** of the posts **410** are received in the lower retaining feature openings **332** and the upper radially outward extending projections **416** at the top end **444** are received in the upper retaining feature openings **332**, i.e., in the lower locking feature openings **334**. In the initial configuration, the radially outward extending projections **416** of the inner member **400** engage with the respective retaining feature openings **332** to prevent downward movement of the inner member **400** relative to the outer member **300**, i.e., removal of the inner member **400** from the outer member **300**.

[0093] Further, in the initial configuration, the scannable feature **434** on the inner member **400** is covered by the shield portion **346** of the outer member **300**.

[0094] In FIG. **12**, the attachment assembly **100** is shown in the initial configuration **100'** positioned over an attachment feature **42** of the vehicle **10**. In the illustrated embodiment, the attachment feature **42** includes a hole **52** extending from a top surface **46** to a bottom surface **48** of a component **44** of the vehicle **10**.

[0095] In FIG. **12**, only a portion of the ring member **200** is illustrated for simplicity. As shown, the ring member **200** is joined to the head **504** of the fastener member **500**.

[0096] In FIG. **12**, the radially outward extending projections **416** of the inner member **400** extend through the outer member **300** and are visible. Further, the bottom ends **412** of the posts **410** extend below the outer member **300** and are visible. The distal end **506** of the fastener member **500** extends further below the bottom ends **412** of the posts **410**.

[0097] As further shown, the shield portion **346** of the outer member **300** covers the scannable feature of the inner member **400** such that the scannable feature is not visible.

[0098] In FIG. **13**, the attachment assembly **100** of FIG. **12** is shown after being brought into initial contact with the surface **46** of the vehicle component **44**. Specifically, the attachment assembly **100** is lowered and the distal end **506** of the fastener member **500** is inserted into the hole **52** until the bottom ends **412** first contact the surface **46**. At this stage, the attachment assembly **100** is still in the initial configuration **100'**.

[0099] As shown, the radially outward extensions **508** may be flexible and may be deflected as the fastener member **500** is inserted into the hole **52**. After passing through the hole **52**, the radially outward extensions **508** extend back to their initial position. As a result, radially outward extensions **508** may contact the bottom surface **48** of the vehicle component **44** to hold the fastener member **500** in the hole **52**. As shown, the radially outward extensions may have a diameter **509** that is greater than the diameter **59** of the hole **52**.

[0100] FIG. **14** illustrates the attachment assembly **100** of FIG. **12** in the attached configuration **100''**, after a downward force has pushed the attachment assembly **100** toward the vehicle component **44**.

[0101] As the attachment assembly **100** is pressed downward after initial contact of the bottom ends **412** of the inner member **400** and the surface **46**, the inner member **400** is stationary while the outer member **300**, ring member **200**, and fastener member **500** move downward. As a result, the sloping surfaces **418** of the radially outward projections **416** cause the radially outward projections

416 to retract within the outer member **300** and out of engagement with the retaining feature openings **332**. The inner surface of the outer member **300** may be formed with channels that limit the radially outward projections **416** to vertical movement.

[0102] The downward force may cause the outer member **300** to continue moving downward until the outer member **300** contacts the surface **46** of the vehicle component **44**. In other embodiments, downward movement of the outer member **300** may stop before contacting the surface **46** of the vehicle component **44**.

[0103] As shown in FIG. **14**, when relative movement between the outer member **300** and inner member **400** is sufficient, the radially outward projections **416** reach the locking feature openings **334** and extend outward, into the openings **334** and through the outer member **300**. Engagement between the radially outward projections **416** and the locking feature openings **334** prevent downward movement of the inner member **400** relative to the outer member **300**. Thus, the attachment assembly **100** is locked in the attached configuration **100''**. Internal surfaces of the outer member **300** may abut the exterior surfaces of the inner member **400** in the attached configuration **100''** to prevent further upward movement of the inner member **400** relative to the outer member **300**.

[0104] In the attached configuration **100''**, the scannable feature **434** is aligned with, and visible through, the window **348** in the outer member **300**.

[0105] The embodiments of FIGS. **2-7** and **8-14** prevent scanning of the scannable feature **434** in the initial configuration **100'** of the attachment assembly **100**, and allow scanning of the scannable feature **434** in the attached configuration **100''** of the attachment assembly **100**. Specifically, in each embodiment, the shield portion of the outer member **300** covers the scannable feature **434** of the inner member **400** in the initial configuration **100'**, and the window in the outer member **300** reveals the scannable feature **434** of the inner member **400** in the attached configuration **100''**.

[0106] Referring now to FIG. **15**, a flow chart of a method **900** for ensuring attachment is illustrated. As shown, method **900** includes, at operation **905**, providing an attachment assembly in an initial configuration.

[0107] Method **900** further includes surrounding a coolant hose with the ring member of the attachment assembly at operation **915**.

[0108] At operation **925**, method **900** includes retaining the attachment assembly in the initial configuration, such as by engaging features of the inner member and outer member of the attachment assembly.

[0109] Method **900** includes, at operation **935**, advancing the attachment assembly in a first direction into contact with an attachment feature such that one component of the attachment assembly abuts the attachment feature while a second component of the attachment assembly continues moving in the first direction relative to the first component until the attachment assembly is in an attached configuration in which the scannable feature is revealed by the window.

[0110] Method **900** may further include locking the attachment assembly in the attached configuration by engaging locking features of the inner member and outer member of the attachment assembly at operation **945**.

[0111] At operation **955**, method **900** includes scanning the scannable feature through the window of the outer member with a scanner to confirm that the attachment assembly has been attached to the attachment feature.

[0112] Method **900** may further include communicating a signal from the scanner indicating that the attachment assembly has been attached to the surface to a control module or processor at operation **965**.

[0113] Embodiments described herein may eliminate use of threaded fastening, reducing the cost of parts. Further, embodiments described herein provide for a high level of detection for proper attachment, as the scannable feature cannot be scanned unless the attachment assembly is in the attached configuration. During manufacturing, a failure to scan all scannable features may result in

a line stoppage to ensure that all attachments are properly performed.

[0114] While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth in the appended claims and the legal equivalents thereof.

Claims

1. A method for ensuring attachment, the method comprising: providing an attachment assembly in an initial configuration, wherein the attachment assembly comprises: a first component having a scannable feature; and a second component having a shield portion and a window, wherein, in the initial configuration the scannable feature is received within the second component and the shield portion covers the scannable feature; advancing the attachment assembly in a first direction into contact with an attachment feature such that the first component abuts the attachment feature while the second component continues moving in the first direction relative to the first component until the attachment assembly is in an attached configuration in which the scannable feature is revealed by the window; and scanning the scannable feature through the window with a scanner to confirm that the attachment assembly has been attached to the attachment feature.
2. The method of claim 1, further comprising communicating a signal from the scanner indicating that the attachment assembly has been attached to the attachment feature.
3. The method of claim 1, wherein the second component includes a body portion connected to a ring portion, and wherein the method further comprises surrounding a hose with the ring portion.
4. The method of claim 1, wherein the second component includes a body portion connected to a ring portion, and wherein the method further comprises surrounding a hose with the ring portion before advancing the attachment assembly in the first direction into contact with the attachment feature and before scanning the scannable feature through the window with the scanner to confirm that the attachment assembly has been attached to the attachment feature.
5. The method of claim 1, wherein: the first component includes a first locking feature; the second component includes a second locking feature; and the method comprises locking the attachment assembly in the attached configuration by engaging the first locking feature and the second locking feature.
6. The method of claim 1, wherein: the first component includes a first retaining feature; the second component includes a second retaining feature; and the method comprises retaining the attachment assembly in the initial configuration by engaging the first retaining feature and the second retaining feature until a force from advancing the attachment assembly in the first direction while the first component abuts the attachment feature causes the first retaining feature and the second retaining feature to disengage.
7. The method of claim 1, wherein: the attachment feature includes a surface formed with a hole; the second component includes a distal projection; advancing the attachment assembly in the first direction includes inserting the distal projection into the hole; advancing the attachment assembly in the first direction is continued until the shield portion abuts the surface; the attachment assembly is in the attached configuration when the shield portion abuts the surface; and while advancing the attachment assembly in the first direction the first component abuts the surface.
8. The method of claim 7, wherein: the surface is a first surface of a wall; the wall has an opposite second surface; the hole extends through the wall; the distal projection includes a deformable

radially-outward extension; advancing the attachment assembly in the first direction comprising deforming the deformable radially-outward extension to pass through the hole; and the deformable radially-outward extension extends outward after passing through the hole and prevents removal of the distal projection from the hole.

9. The method of claim 1, wherein: the attachment feature includes a stud extending from a surface; the second component includes an inward-extending arm; advancing the attachment assembly in the first direction includes contacting the inward-extending arm with the stud; advancing the attachment assembly in the first direction is continued until the shield portion abuts the surface; and the attachment assembly is in the attached configuration when the shield portion abuts the surface.

10. The method of claim 9 wherein: the inward-extending arm comprises a plurality of inward-extending arms that define a central passageway; and while advancing the attachment assembly in the first direction a distal portion of the stud extends through the central passageway.

11. An attachment assembly comprising: a first component having a scannable feature; and a second component having a shield portion and a window: wherein: the first component is movable relative to the second component in a first direction from an initial configuration to an attached configuration; in the initial configuration the scannable feature is received within the second component and the shield portion covers the scannable feature; and in the attached configuration, the scannable feature is revealed by the window.

12. The attachment assembly of claim 11, wherein: the first component includes a first locking feature; the second component includes a second locking feature; and the first locking feature is configured to engage the second locking feature to lock the attachment assembly in the attached configuration.

13. The attachment assembly of claim 11, wherein: the first component includes a first retaining feature; the second component includes a second retaining feature; and the first retaining feature is configured to hold the second retaining feature in the initial configuration and to release the second retaining feature upon application of a force in the first direction.

14. The attachment assembly of claim 11, wherein: the second component includes a distal projection extending in a second direction opposite the first direction and configured for insertion in a hole in a wall; and the distal projection includes a deformable radially-outward extension configured to engage a backside of the wall.

15. The attachment assembly of claim 11, wherein the second component includes inward-extending arms that terminate at a central passageway.

16. A vehicle comprising: a coolant system including a hose; and an attachment assembly comprising: a first component having a scannable feature; and a second component including a ring portion engaging the hose and having a shield portion and a window; wherein: the first component is movable relative to the second component in a first direction from an initial configuration to an attached configuration; in the initial configuration the scannable feature is received within the second component and the shield portion covers the scannable feature; and in the attached configuration, the scannable feature is revealed by the window.

17. The vehicle of claim 16, further comprising a wall formed with a hole extending from a first surface of the wall to an opposite second surface of the wall, wherein: the second component includes a distal projection extending through the hole; the distal projection includes a radially-outward extension having a diameter greater than a diameter of the hole; the shield portion abuts the first surface of the wall; and the first component abuts the first surface of the wall.

18. The vehicle of claim 16, further comprising a stud extending from a surface; wherein the second component includes inward-extending arms that terminate at a central passageway; the stud passes through the central passageway; and the first component abuts the surface.

19. The vehicle of claim 16, wherein: the first component includes a first locking feature; the second component includes a second locking feature; and the first locking feature is configured to engage the second locking feature to lock the attachment assembly in the attached configuration.

20. The vehicle of claim 16, wherein: the first component includes a first retaining feature; the second component includes a second retaining feature; and the first retaining feature is configured to hold the second retaining feature in the initial configuration and to release the second retaining feature upon application of a force in the first direction.
